



TEMASEK
JUNIOR COLLEGE

Preliminary Examinations
Higher 1

CANDIDATE
NAME

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CIVICS
GROUP

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CHEMISTRY
Paper 2

8872/02
17th September 2013

Candidates answer Section A on the Question Paper.

2 hours

Additional Materials: Answer Paper
Data Booklet
Graph Paper

READ THESE INSTRUCTIONS FIRST

Write your Civics Group number and candidate name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **two** questions on separate answer paper.

At the end of the examination, Section A and
Section B will be handed in separately.
The number of marks is given in brackets [] at the
end of each question or part question.

For Examiner's Use		
Paper 1	/30	
Paper 2	A1	/22
	A2	/8
	A3	/10
	B	/20
	B	/20
Total		

Section A

Answer **all** questions in this section in the spaces provided.

- 1** This question is about ethanol and its metabolism in human body.

Statistics show a correlation between road accidents and drivers who drink too much ethanol-based beverages. One of the first chemical breathalysers used by the Singapore Traffic Police was based on the oxidation process of ethanol using acidified potassium dichromate(VI).

- (a) (i) Derive the ion-electron half equation for the oxidation of ethanol to ethanoic acid.

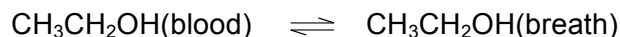
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- (ii) Using relevant information from the *Data Booklet* and the ion-electron half equation derived in (i), write a balanced ionic equation for the oxidation of ethanol using acidified potassium dichromate(VI).

.....

A breathalyser kit consists of an inflated 1000 cm³ plastic bag connected to a glass tube containing dichromate(VI) crystals. When breath is expelled through the tube, the crystals changed colour as they are reduced. The proportion of the crystals that changed colour indicated the amount of ethanol present.

The current legal maximum blood ethanol concentration when driving is 80 mg per 100 cm³ of blood. Ethanol concentration in the blood can be estimated by analysing the ethanol content in breath because an equilibrium is set up between the ethanol in the blood and the ethanol in breath:



At body temperature, the concentration of ethanol in the blood is about 2300 times that in the breath.

- (iii) Write an expression for K_c for the equilibrium of ethanol between the blood and the air in the lungs.

(iv) Calculate a value for K_c .

(v) In Singapore, what is the legal maximum ethanol concentration in breath in mg per 1000 cm^3 ?

(vi) What is the ethanol concentration calculated in (v) in mol dm^{-3} ?

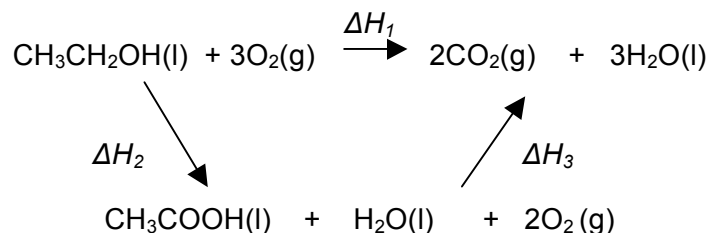
(vii) What colour change would you expect for a positive result in the breathalyser?

.....

[9]

- (b) Ethanol can be burnt in excess oxygen as a fuel but in blood, ethanol is partially oxidised. Complete combustion of ethanol releases 1367 kJ mol^{-1} of energy while the metabolism of ethanol in blood leads to partial oxidation which releases 770 kJ mol^{-1} of energy. Complete combustion of ethanoic acid, CH_3COOH , releases 873 kJ mol^{-1} of energy.

The diagram below shows the energy cycle involving ethanol.



- (i) State Hess' Law.

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- (ii) Write an equation that links ΔH_1 , ΔH_2 and ΔH_3 .

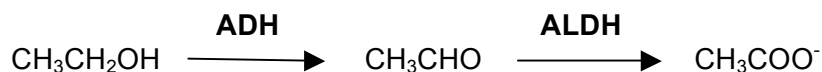
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- (iii) Use the energy cycle above to calculate ΔH_2 , the energy released if 1.0 mol of ethanol was to be oxidised to ethanoic acid and water.

- (iv) Assuming that in human blood, ethanol is oxidised to a mixture of ethanoic acid and carbon dioxide, use the data given to calculate the proportion of ethanol metabolised to each substance.

[7]

- (c) Most of the ethanol in the body is broken down by an enzyme (biological catalyst) called alcohol dehydrogenase (ADH), which transforms ethanol into an intermediate compound called ethanal (CH_3CHO), a known carcinogen. However, ethanal is generally short-lived; it is quickly broken down to a less toxic compound called ethanoate (CH_3COO^-) by another enzyme called aldehyde dehydrogenase (ALDH). The overall reaction is accompanied by the release of energy.



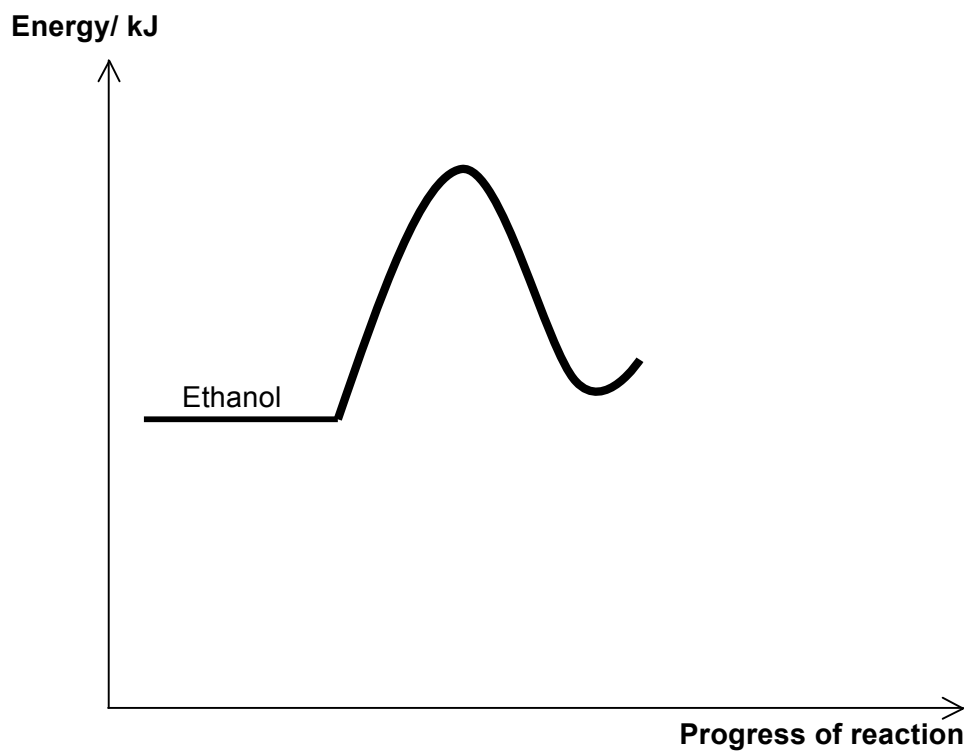
- (i) Define the term 'catalyst'.

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.....

.....

- (ii) Complete the reaction pathway diagram below for the breakdown of ethanol to ethanoate ion, showing clearly any intermediate formed in the reactions.



- (iii) Explain clearly why ethanoate ion is soluble in blood plasma assuming the main composition in blood plasma is water.

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[6]

[Total: 22]

- 2** The elements in the third period of the Periodic Table show a change from metallic to non-metallic nature, which is reflected in the bonding and reaction of their chlorides.

(a) State the formulae for chlorides of aluminium and silicon.

..... **[1]**

(b) Describe briefly the reaction of aluminium and silicon when each is reacted with chlorine.

aluminium

.....
.....
.....

silicon

.....
.....
.....

[2]

- (c) Using the chlorides of aluminium and silicon, describe their reactions with water. Write equations where appropriate and describe the effect of adding Universal Indicator to the resulting solution.

aluminium chloride

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Universal Indicator turned

silicon chloride

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Universal Indicator turned

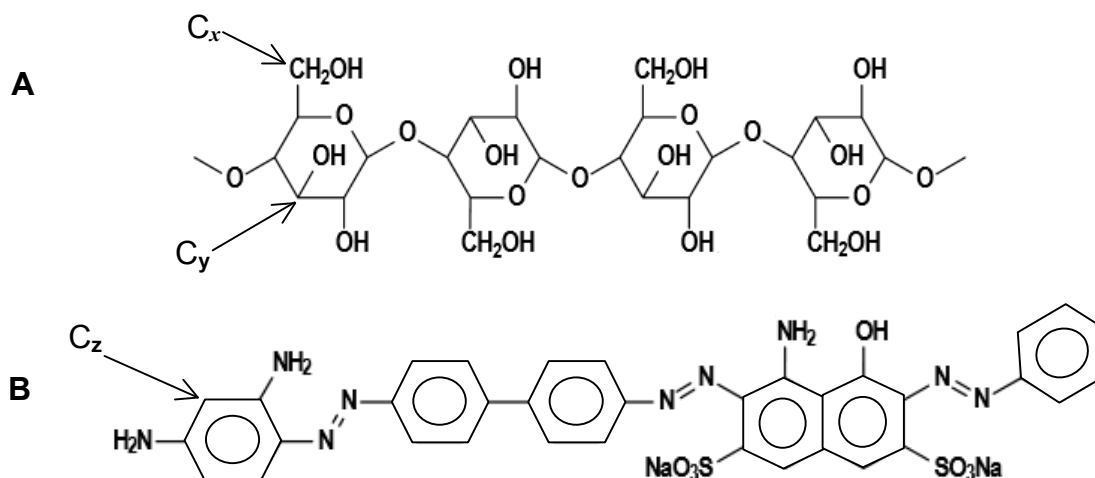
[5]

[Total:8]

- 3 Cellulosic materials include cotton, rayon, and linen, all of which are very polar and hydrophilic, and therefore, require polar, water-soluble dyes for their colouration from a dyebath. In addition, dyes must be designed that maintain affinity when the cellulosic substrate is exposed to water.

Direct dyes were the first water-soluble colourants that had affinity for cotton in the absence of a binding agent.

(a) The diagram below shows a fragment of a cellulose, **A**, and a direct dye **B**.



- (i) Write down the hybridised state of each of the carbon atoms labelled C_x , C_y and C_z above.

C_x

C_y

C_z

- (ii) Name the functional groups that are present in cellulose, **A**.
You may ignore the $-C-O-C-$ functional groups.

.....

- (iii) Hydrogen bonding was found to exist between cellulose, **A**, and direct dye, **B**.

Draw on the diagram above to show how **two** such bonds occur.

[5]

(b) 2,4–dinitrophenylhydrazine is a useful reagent in synthesising organic dyes. When a carbonyl compound reacts with this reagent, a coloured hydrazone is formed.

(i) Suggest the colour of the hydrazone.

.....

(ii) Name the type of reaction that has taken place.

.....

(iii) In a laboratory at room temperature, a student reacted ethanal with 2,4–dinitrophenylhydrazine. Write a balanced equation for the reaction that has occurred.

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(iv) The organic product from the reaction in (iii) exists as two geometric isomers. Explain how this is possible.

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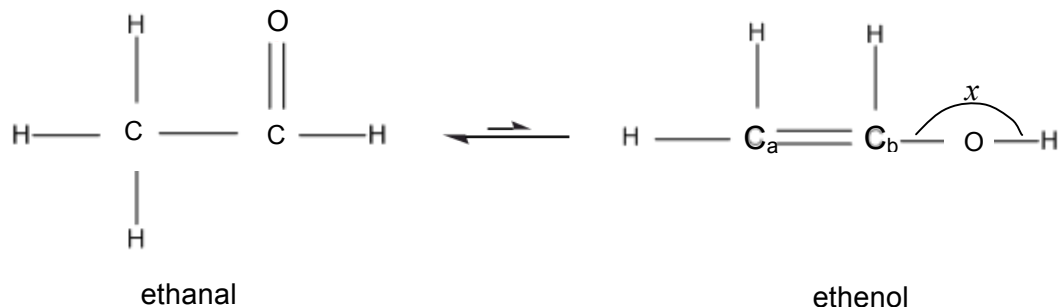
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[5]
[Total: 10]

Section B

Answer **two** of the three questions in this section on separate paper

- 4 (a) Vinyl alcohol, also called ethenol (IUPAC name), is an alcohol with molecular formula, CH_2CHOH . At room conditions, it converts favourably to give the ethanal form.



- (i) State the oxidation numbers for C_a and C_b in ethenol.
- (ii) State the Valence Shell Electron Pair Repulsion (VSEPR) Theory.
- (iii) By considering the number of lone and bond pairs of electrons, use the VSEPR Theory to state the shape with respect to the oxygen atom in ethenol. Hence, suggest a value for the bond angle, x , labelled above.
- (iv) Vinyl alcohol can be used to synthesise polyvinyl alcohol, which is commonly used in eye drops and hard contact lens solution as a lubricant. The repeat unit of polyvinyl alcohol is given below.



Research studies have shown that esterification of polyvinyl alcohol with stearic acid leads to better thermal stability and water resistance of the compound. Using $\text{R}-\text{CO}_2\text{H}$ to represent the stearic acid molecule, write down the repeat unit for the polyester obtained.

[8]

- (b) Red lead primer paint, used in metal painting contains red lead oxide, which behaves chemically as a mixture of lead(II) oxide, PbO , and lead(IV) oxide, PbO_2 .

A 2.50 g sample of solid red lead oxide was mixed with an excess of acidified potassium iodide solution. PbO_2 reacted to give $\text{Pb}^{2+}(\text{aq})$ and $\text{I}_2(\text{aq})$.

- Write a balanced ionic equation, including state symbols, for the reaction of PbO_2 with acidified potassium iodide solution.
- Upon completion of the reaction, the resulting mixture was diluted to 100 cm^3 with water. A 10.0 cm^3 portion of the solution was withdrawn and then titrated with 12.50 cm^3 of $0.050 \text{ mol dm}^{-3}$ sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$ solution.

Calculate the number of moles of I_2 produced from the reaction of PbO_2 with acidified potassium iodide solution.

- Hence, calculate the percentage by mass of PbO_2 in the red lead oxide sample.

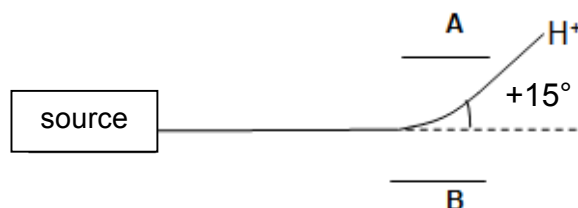
[5]

- (c) An unknown element, **X**, is found to have the electronic configuration of $[\text{Ar}] 3d^6 4s^1 5s^1$ when it is in the excited state.

- Suggest the identity of **X**.
- Draw the energy level diagram showing the electronic configuration of **X** in the ground state.

Beams of charged particles are deflected by an electric field. If the particles are all travelling at the same speed, through an electric field of constant strength, the angle of deflection is proportional to their charge/mass ratio.

In a particular experimental set-up, protons are deflected through an angle of $+15^\circ$. The path of deflection is shown in the diagram below:

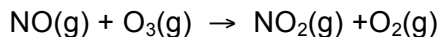


- X** forms X^{2+} after being bombarded by high energy electrons. Write the electronic configuration of X^{2+} in its ground state.
- State the polarity (+ or –) of plate **A**.
- Estimate the angle of deflection when a beam of X^{2+} is subjected to the electric field. Explain your answer.

[7]

[Total: 20]

- 5 In the atmosphere, nitrogen monoxide reacts with ozone as shown in the following equation.



- (a) Briefly explain how NO is formed in an internal combustion engine.
- (b) NO₂ is considered to be an air pollutant. State one undesirable consequence of the presence of NO₂ in the atmosphere.
- (c) Both NO and O₃ at an initial concentration of $1.0 \times 10^{-6} \text{ mol dm}^{-3}$ were mixed together at 25 °C.

A student calculated the following values using $\frac{\Delta[\text{NO}_2]}{\Delta t}$, [NO] and [O₃] at each timing as shown in the Table below.

Time / s	$\frac{\Delta[\text{NO}_2]}{\Delta t}$	[NO] / mol dm ⁻³	[O ₃] / mol dm ⁻³
0.05	1.00×10^{-5}	5.00×10^{-7}	5.00×10^{-7}
0.15	1.60×10^{-6}	2.30×10^{-7}	2.30×10^{-7}
0.25	6.00×10^{-7}	1.50×10^{-7}	1.50×10^{-7}
0.30	4.00×10^{-7}	1.30×10^{-7}	1.30×10^{-7}

- (i) What does $\frac{\Delta[\text{NO}_2]}{\Delta t}$ represent? State its units.
- (ii) It has been found that the following is the rate equation for the reaction of NO with O₃.

$$\text{Rate} = k[\text{NO}][\text{O}_3]$$

The reaction is overall second order.

What is understood by the term *overall order of reaction*?

- (iii) Use the information given in the table above to plot a straight line graph to confirm that the reaction shows overall second order kinetics. Hence, determine the rate constant, k .
- (iv) The rate of reaction was found to increase with increasing temperature. With the aid of a suitable diagram, explain why an increase in temperature could cause an increase in the rate of reaction.

[12]

- (d) Compound **K** has the molecular formula $C_4H_8O_2$. When 0.1 mol of **K** is reacted with an excess of sodium metal at room temperature and pressure, 1.2 dm^3 of hydrogen are formed. Compound **K** does not react with aqueous sodium hydroxide. **K** forms a pale yellow precipitate when warmed with alkaline aqueous iodine. A silver mirror is observed when **K** is warmed with Tollen's reagent.

Compound **L** an isomer of **K**, is neutral to litmus and has a sweet smell. On treatment with hot aqueous sodium hydroxide, **L** yields **M** $C_2H_3O_2Na$ and **N** C_2H_6O .

Deduce with reasoning, the structures of **K**, **L**, **M** and **N**.

[8]

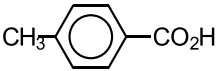
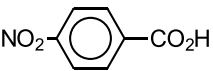
[Total: 20]

6 This question is about acids.

- (a) (i) Propanoic acid can be synthesised from bromoethane. Suggest a synthetic pathway, draw the intermediates and state the reagents and conditions.
- (ii) Predict and explain the effect on the rate of reaction if chloroethane is used instead as the starting material for the synthesis of propanoic acid.

[5]

- (b) Benzoic acid is a carboxylic acid that occurs naturally in many plant and animal species. Substituents on aromatic rings affect the acidity of substituted benzoic acids.

The pH of 1 mol dm^{-3} of 4-methylbenzoic acid, , and 4-nitrobenzoic acid, , (not necessarily in that order) were measured and the following data was obtained:

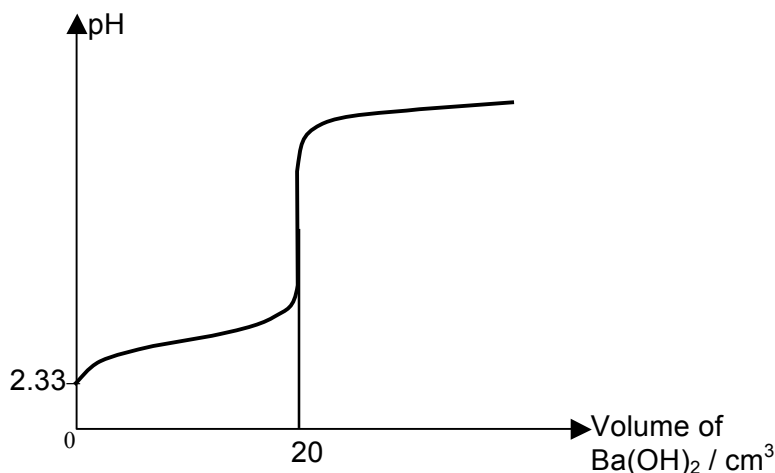
Solution	pH
1	1.71
2	2.17

Given that the pH of 1 mol dm^{-3} of benzoic acid is 2.10, deduce the identities of the solutions, explain your answer.

[3]

- (c) Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$, is produced in muscle cells during exercise and is being converted into energy by *mitochondria*, the energy factories in cells. By determining how effective the muscle *mitochondria* takes up the lactic acid produced, sports scientists can help athletes optimise their training regime.

A sample of blood is taken from the athlete and the lactic acid content is analysed. 25 cm^3 of lactic acid is titrated with $0.100 \text{ mol dm}^{-3}$ of $\text{Ba}(\text{OH})_2$ and the titration curve is shown below.



- (i) Determine the concentration of the lactic acid.
- (ii) Hence, verify that lactic acid is a weak acid.
- (iii) Write an equation to show the dissociation of lactic acid and calculate the K_a of lactic acid.
- (iv) Given the following data,

Indicator	Working pH range
Bromocresol green	3.8 to 5.5
Thymol blue	8.0 to 9.6

Suggest the indicator that is suitable for the above titration, explain your reasoning. [7]

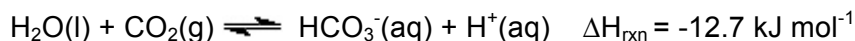
- (d) Strenuous exercise could lower the blood pH resulting in a condition known as acidosis. Blood pH is maintained at a range of 7.35 to 7.45 by a complex system of buffers. The main buffering agent in blood plasma is the carbon dioxide/hydrogencarbonate system:



- (i) What is meant by the term *buffer*?
- (ii) Explain how the above buffer system works when OH^- ions are added and write relevant equations.
- (iii) Besides metabolic processes, respiratory processes could also lead to changes in blood pH. For instance, rapid breathing that occurs during anxiety could cause the carbon dioxide levels in blood to drop and the blood pH to be higher than the optimal range.

One of the ways to treat this condition is to breathe into a bag to “re-breathe” the exhaled carbon dioxide. Suggest how this can bring the blood pH to the optimal range.

- (iv) Given the following data,



Standard enthalpy change of combustion of C / kJ mol^{-1}	-393.5
Standard enthalpy change of combustion of H_2 / kJ mol^{-1}	-285.8
Standard enthalpy change of formation of H^+ / kJ mol^{-1}	0

Calculate the standard enthalpy change of formation of $\text{HCO}_3^-(\text{aq})$.

[5]

[Total: 20]

